



Guideline

High Output Stoma

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Hospital-Wide	Medical, Nursing, Dietetics Pharmacy

Disclaimer: This document does not override decision based on clinical judgement and experience of the prescriber.

Introduction

Patients with a high output stoma (HOS) are at risk of developing complications including dehydration, renal impairment, malnutrition, electrolyte abnormalities and long-term gut dysfunction. These complications can lead to increased length of stay and reduced quality of life. This guideline aims to ensure that HOS patients receive nutritional and medication management in line with evidence-based practice in order to meet their nutrition and hydration needs post-surgery and improve patient outcomes.

The output of a stoma depends on the volume of food and fluids consumed, and the volume of gastrointestinal secretions. Early high output from an ileostomy is common and 49% resolve spontaneously¹.

This guideline has been developed to aid consistent management of these inpatients at SCGOPHCG by providing a decision support tool for the clinical workforce. The guideline applies to jejunostomies, ileostomies and colostomies.

Clinical Presentation

High Output Stoma	- Greater than 1500 mL of watery stool over 24 hours for 3 consecutive days. - Signs of dehydration - Electrolyte disturbance (sodium, magnesium, potassium, phosphate), elevated creatinine, reduced glomerular filtration rate (GFR) - Unintentional loss of weight
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Management Strategies

Refer to the Medical Team, Stomal Therapy Nurse, Dietitian and Pharmacist

Exclude potential causes by ruling out the following:

- Intra-abdominal sepsis
- Partial or intermittent bowel obstruction
- Enteritis e.g. *Clostridium difficile*
- Sudden drug withdrawal e.g. steroids or opioids
- Administration of prokinetics e.g. metoclopramide, domperidone
- Recurrent disease e.g. Crohn's Disease
- Bacterial overgrowth from diverticular or blind loop fermentation

Stage 1. Reduce fluid and electrolyte losses

- Rehydrate with intravenous hydration (IVH). 'For patients with associated hyponatraemia, avoid fluid restriction as first line treatment'. Provide normal saline IVH.
- Restrict normal (hypotonic and hypertonic) oral fluids to 500 mL daily. This includes water, tea, coffee, nutritional supplement drinks, fruit juice, cordial, milk, soft drinks and so on. Nursing staff to update the Automated Menu System with 500mL fluid restriction.
- Maintain a strict fluid balance chart.
- Monitor daily weight.
- Monitor daily urea and electrolytes (U&E), magnesium and phosphate.
- Commence loperamide 4 mg QID to reduce stoma losses. Capsules should be opened, mixed with a small amount of water or food and given 30 minutes prior to meals and at bedtime.
- Commence proton pump inhibitor (PPI) – pantoprazole 40mg daily to reduce gastric secretions.
- If serum magnesium is less than 0.5 mmol/L replace with 10 to 20 mmol of magnesium sulphate IV. Each 10 mmol/L in 100 mL 0.9% sodium chloride infused over at least ONE hour.
- Encourage patients to eat "Regular" hospital diet.

Stage 2. Optimise treatment with antisecretory/diarrhoeal medication

- Continue 500 mL normal (hypotonic and hypertonic) oral fluid restriction. This includes water, tea, coffee, nutritional supplement drinks, fruit juice, cordial, milk, soft drinks and so on.
- In addition to oral fluid restriction commence an oral rehydration solution. Start with 1000 mL daily and sip slowly over the day between meals. Nursing staff to label solution with details of the contents and date of reconstitution using miscellaneous route label (158732C). Discard after 24 hours. Medical staff to chart on WA Hospital Medication Chart (WA HMC) daily at 0800hrs:
 - Oral rehydration salts (ORS)® - 8 sachets in 1000 mL water

If ORS® is unavailable, contact Pharmacy for suitable alternative. Suitable alternatives are:

- Hydralyte™ effervescent tablets – 15 tablets in 1000 mL water, OR
- APOHEALTH™ effervescent tablets – 15 tablets in 1000 mL water.

Please note:

- Oral rehydration solutions are a glucose-sodium fluid that are made up to approximately 90 mmol/L of sodium. This means that ORS®, Hydralyte™ and APOHEALTH™ are made up to 1 ½ strength and not as per Manufacturer's instructions.
- Do NOT add ice or other liquids to made up oral rehydration solution as this will dilute the concentration.

- If the stoma output is greater than > 3000 mL daily, consider placing the patient nil by mouth (NBM) for 24 hours to assess gastrointestinal secretion.
- Continue IVH as required.
- Continue PPI.
- Increase loperamide to 8 mg QID to reduce stoma losses. Capsules should be opened, mixed with a small amount of water or food and given 30 minutes prior to meals and at bedtime.
- Continue daily monitoring of urea and electrolytes (U&E), magnesium and phosphate, and replace electrolytes as required.

Stage 3. Evaluate efficiency of additional treatment if HOS continues

- Increase loperamide by 2 to 4 mg QID to reduce stoma losses. Capsules should be opened, mixed with a small amount of water or food and given 30 minutes prior to meals and at bedtime. (Do not exceed 8 mg QID without specialist consultation). In certain circumstances it may be appropriate to exceed dose with consultation from consultant. Consider ECG monitoring if using high doses of loperamide, particularly in patients with history of cardiac conditions.
- Addition of codeine phosphate 15 to 30 mg QID, 30 to 60 minutes before meals. Use with caution in renal/hepatic impairment, and the elderly as codeine is addictive and sedating
- Review compliance to oral fluids and increase volume of oral rehydration solution if required.
- Consider commencing octreotide 100 to 200 micrograms TDS for 3 to 5 days used under specialist consultation, as per Formulary One restrictions. If there is no significant improvement in stoma output discontinue use.
- Dietitian to assess for dietary fibre modification, as required.

Stage 4. Specialised Nutrition and Electrolyte Replacement

- If the patient has had four days of high stoma output with the above treatment (Stage 3) refer patient for Total Parenteral Nutrition (TPN) and refer to the Colorectal team for specialised review.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising Policy and Standard/s:

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- NSQHS Standard 1,3,4,5,6

Standards, procedures, guidelines:

- [SCGOPHCG Nursing Practice Guidelines No.41 Colostomy Ileostomy Urostomy](#)

Supporting documents

- SCGOPHCG Fluid Balance Chart MR904 MR(OPH)144
- Stoma Observation and Education Care Plan SCGH MR872/ OPHMR124.1

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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